control and high-bandwidth sensor streams, and anti-jamming/crypto modules for secure links.

- *Mission computer and onboard processing*: Ruggedized high-performance processors/GPUs for real-time sensor processing, machine-learning inference, data fusion, and health monitoring. These execute autonomy, ATR, flight-path replanning, and compressed data products.
- *Sensor suite/payloads*: Typical combinations include electro-optical/infrared (EO/IR) gimbals, synthetic aperture radar (SAR) with ground moving target indication (GMTI), signals intelligence (SIGINT), electronic warfare (EW) packages, LiDAR, hyperspectral imagers, and maritime sensors (AIS receivers).
- *Weapons and effectors (if applicable)*: For armed platforms: precision weapons interfaces (e.g., laser-guided missiles), integrated fire-control and weapon release mechanisms complying with safety interlocks and mission rules of engagement.
- *Power and energy management*: Onboard fuel systems or battery packs with power management units that service propulsion, avionics, and payloads; energy budgeting software optimizes sensor use to conserve endurance.
- *Launch and recovery systems depending on platform*: Conventional runway, parachute/recovery nets, autonomous arrested recovery, or autonomous carrier operations for naval UCAVs.

4.3.3 Major software and systems components

- *Autonomy stack*: Flight management, path planning, obstacle avoidance, and mission logic layers. Increasingly includes ML models for perception and decision support.